

**In one line:** A **macro** (Swift 5.9) is a compiler plugin that receives the **syntax tree** of its use site and returns **new syntax**, spliced in at compile time — *additive only*, no runtime cost. A **result builder** (`@resultBuilder`, Swift 5.4) rewrites the statements of a closure into nested static `build...` calls that fold them into *one* value — the engine behind SwiftUI's `@ViewBuilder`.

## Macros

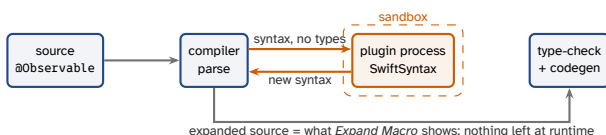
- **Declaration** (your library): `macro stringify<T>(_ v: T) -> (T, String) = #externalMacro(module: "MyMacros", type: "StringifyMacro")`. **Implementation:** a `macro target` (SwiftPM) — structs conforming to `ExpressionMacro`, `MemberMacro`, ... with a static expansion(`of:..in:`), registered by an `@main CompilerPlugin`.
- The plugin runs as a **separate, sandboxed process** (no file system, no network): input = syntax nodes (SwiftSyntax), output = syntax. It sees **no types** — only the text of the code.
- **Hygienic:** invented names come from `makeUniqueName()`; names your code will use must be declared (names: `named(x)`, `prefixed(_)`, `arbitrary`). Errors via `context.diagnose(...)`.
- **Debug:** right-click → **Expand Macro** in Xcode; test with `assertMacroExpansion` (SwiftSyntaxMacrosTestSupport). Package macros need *Trust & Enable* (CI: `xcodebuild -skipMacroValidation`).

| Role                                                        | Adds                                                                     | Real example                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Freestanding #name</b> — <code>@freestanding(...)</code> | expression<br>declaration                                                | one value<br>decis at that spot<br><br><code>#expect(a == b)</code><br><code>#Preview { V() }</code>                                                                                                                                                                                                                                                                             |
| <b>Attached @Name</b> — <code>@attached(...)</code>         | peer<br>member<br>accessor<br>memberAttribute<br>extension<br>body (6.0) | decis <i>beside</i> the target<br>decis <i>inside</i> the type<br>get/set: stored → computed<br>attributes on each member<br>extension + conformances<br>a function body<br><br><code>@Test func t()</code><br><code>@Observable registrar</code><br><code>@ObservationTracked</code><br><code>@Observable → tracked</code><br><code>: Observable</code><br><code>SE-0415</code> |

## Example — @Observable, expanded (abridged)

```
@Observable final class Model { var name = "" }
// -> Expand Macro:
final class Model {
    @ObservationTracked var name = "" // memberAttribute
    @ObservationIgnored private let _$observationRegistrar =
        Observation.ObservationRegistrar() // member
    internal nonisolated func access<M>(keyPath: KeyPath<Model, M>)
    internal nonisolated func withMutation<M, R>(...) // member
}
extension Model: Observation.Observable {} // extension
// @ObservationTracked (accessor + peer) turns 'name' into
// get { access(keyPath: \.name); return _name }
// set { withMutation(keyPath: \.name) { _name = newValue } }
// + peer storage @ObservationIgnored private var _name = ""
```

## Where a macro runs



## Macro traps

- ! Macros **cannot see types** (`#m(x)` gets the text "x", not `Int`) and **cannot delete or rewrite** the code they annotate — they only add.
- ! Names not declared in `names:` are invisible → "cannot find in scope".

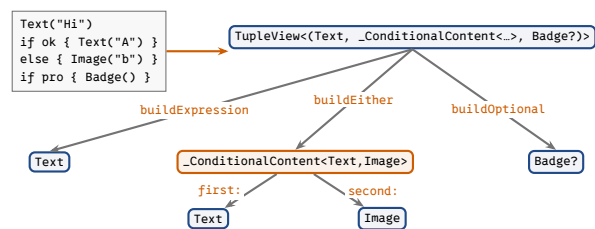
## Result builders

Mark a type `@resultBuilder`; put it on a closure parameter or computed property (`@ViewBuilder var body`). The compiler **rewrites** the body at compile time; the `build...` methods **run at runtime** with real values. Each statement kind needs its method:

| Method                                                      | Enables                                |
|-------------------------------------------------------------|----------------------------------------|
| <code>buildBlock(_ c: C...) -&gt; C</code>                  | a block — the <b>only required</b> one |
| <code>buildExpression(_ e: E) -&gt; C</code>                | lifts each line; overload per input    |
| <code>buildOptional(_ c: C?) -&gt; C</code>                 | if without else, if let                |
| <code>buildEither(first:) / (second:)</code>                | if/else, switch                        |
| <code>buildArray(_ cs: [C]) -&gt; C</code>                  | for...in                               |
| <code>buildLimitedAvailability</code>                       | if #available (erases type)            |
| <code>buildPartialBlock(first:), (accumulated:next:)</code> | pairwise fold, any count (5.7)         |
| <code>buildFinalResult</code>                               | convert to the public type             |

```
@resultBuilder enum Lines {
    static func buildBlock(_ p: String...) -> String {
        p.joined(separator: "\n")
    }
    static func buildOptional(_ p: String?) -> String {
        if let p { p } else { "" }
    }
}
func doc(@Lines _ make: () -> String) -> String { make() }
let s = doc { "title"; if vip { "gold" } }
// = buildBlock("title", buildOptional(vip ? "gold" : nil))
```

## What @ViewBuilder builds



The two branches are **different views** (structural identity): flipping `ok` destroys `A`'s `@State` and runs a transition. Keep identity with a modifier: `.opacity(ok ? 1 : 0)`.

## Builder traps

- ! `ViewBuilder` has **no** `buildArray`: a `for` in body does not compile — use `ForEach` (it also carries per-row identity).
- ! An explicit return **turns the transform off**. Not allowed: `while`, `repeat`, `guard`, `break/continue`, `defer`, `do-catch`.
- ! "Max 10 views" came from `buildBlock` overloads — gone with parameter packs (Swift 5.9).

## Remember

**Macro** = syntax in, syntax out, before types exist. **Builder** = one method per statement kind — `Block`, `Expression`, `Optional`, `Either`, `Array`.